FIT3080 Artificial Intelligence
Solutions to Practice Questions in Knowledge Representation

1. [**Entailment**] Prove the following lemma: $\alpha \models \beta$ iff $\alpha \wedge \neg\beta$ is unsatisfiable.

   SOLUTION:

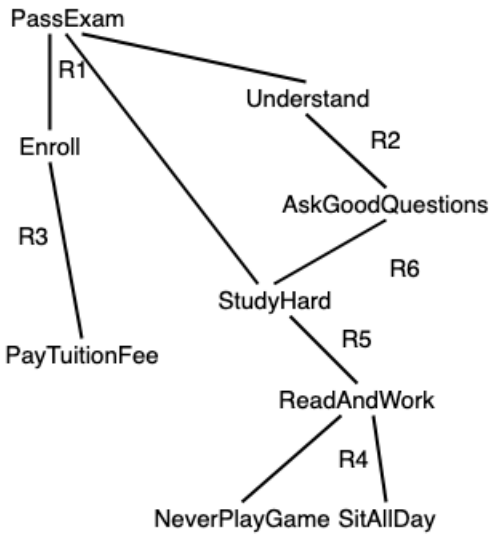
   - $\alpha \wedge \neg\beta \equiv \neg\neg(\alpha \wedge \neg\beta) \equiv \neg(\neg\alpha \vee \beta) \equiv \neg(\alpha \Rightarrow \beta)$
   - According to the following lemma (slide 22), α is valid iff $\neg\alpha$ is unsatisfiable. Therefore $\neg(\alpha \Rightarrow \beta)$ is unsatisfiable iff $\alpha \Rightarrow \beta$ is valid.
   - According to the Deduction Theorem (slide 21), $\alpha \Rightarrow \beta$ is valid iff $\alpha \models \beta$.

2. [**Propositional logic**] Say whether each of the following sentences is syntactically legal in Propositional Logic, and if not, why:

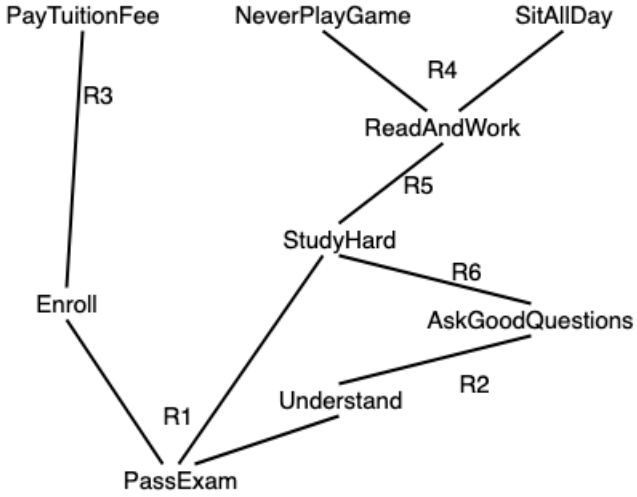
   - $P \wedge \neg P$
     Yes.
   - $\neg(Q \vee R)\neg Q \Rightarrow \neg\neg P$
     No, as there is no connective before $\neg Q$.

3. [**Backward/Forward chaining**] Apply backward and forward reasoning to the following Horn clauses to prove PassExam.
   R1: $Enroll \wedge StudyHard \wedge Understand \Rightarrow PassExam$
   R2: $AskGoodQuestions \Rightarrow Understand$
   R3: $PayTuitionFee \Rightarrow Enroll$
   R4: $NeverPlayGame \wedge SitAllDay \Rightarrow ReadAndWork$
   R5: $ReadAndWork \Rightarrow StudyHard$
   R6: $StudyHard \Rightarrow AskGoodQuestions$
   $PayTuitionFee$
   $NeverPlayGame$
   $SitAllDay$

   SOLUTION:
   Backward chaining:



Forward chaining:



Alias	FullName
PTF	PayTutionFee
NPG	NeverPlayGame
SAD	SitAllDay
E	Enroll
SH	StudyHard
U	Understand
RAD	ReadAndWork
AGQ	AskGoodQuestion
PE	PassExam

Queue	R1	R2	R3	R4	R5	R6	Inferred
PTF, NPG, SAD	3	1	1	2	1	1	
NPG, SAD, E	3	1	0	2	1	1	PTF
SAD, E	3	1	0	1	1	1	PTF, NPG
RAW, E	3	1	0	0	1	1	PTF, NPG, SAD
SH, E	3	1	0	0	0	1	PTF, NPG, SAD, RAW
AGQ, E	2	1	0	0	0	0	PTF, NPG, SAD, RAW, SH
U, E	2	0	0	0	0	0	PTF, NPG, SAD, RAW, SH, AGQ
E	1	0	0	0	0	0	PTF, NPG, SAD, RAW, SH, AGQ, U
PE	0	0	0	0	0	0	PTF, NPG, SAD, RAW, SH, AGQ, U, E

4. [**First Order logic**] Using the predicates PERSON and HAS-PET, represent the following sentence in First Order Logic:
Some people don't have a pet.

   SOLUTION: $\exists x$ [PERSON(x) $\wedge \neg$HAS-PET(x)

5. [**First Order logic**] Assuming predicates $PARENT(p, q)$ and $FEMALE(p)$ and constants $Joan$ and $Kevin$, with the obvious meanings, express each of the following sentences in First Order Logic. (You may use the abbreviation $\exists^1$ to mean "there exists exactly one.")

   (a) Joan has a daughter (possibly more than one, and possibly sons as well).
   SOLUTION: $\exists x\ PARENT(Joan, x) \wedge FEMALE(x)$

   (b) Joan has exactly one daughter (but may have sons as well).
   SOLUTION: $\exists^1\ PARENT(Joan, x) \wedge FEMALE(x)$

   (c) Joan has exactly one child, a daughter.
   SOLUTION:
   $\exists^1 x\ PARENT(Joan, x) \wedge FEMALE(x) \wedge [\forall y\ PARENT(Joan, y) \Rightarrow y = x]$

   (d) Joan and Kevin have exactly one child together.
   SOLUTION: $\exists^1 c\ PARENT(Joan, c) \wedge PARENT(Kevin, c)$

   (e) Joan has at least one child with Kevin, and no children with anyone else.
   SOLUTION: $\exists c\ PARENT(Joan, c) \wedge PARENT(Kevin, c) \wedge$
   $\forall d \forall p\ [PARENT(Joan, d) \wedge PARENT(p, d)] \Rightarrow [p = Kevin]$

6. [**First Order Logic**] Consider a vocabulary with the following symbols:
$OCC(p, o)$: Predicate. Person p has occupation o.
$CUST(p1, p2)$: Predicate. Person $p1$ is a customer of person $p2$.
$BOSS(p1, p2)$: Predicate. Person $p1$ is a boss of person $p2$.
$Doctor, Surgeon, Lawyer, Actor$: Constants denoting occupation.
$Emily, Joe$: Constants denoting people.

Use these symbols to write the following assertions in First Order Logic:

(a) Emily is a surgeon or a lawyer.
SOLUTION: $OCC(Emily, Surgeon) \vee OCC(Emily, Lawyer)$

(b) Joe is an actor, but he also holds another job.
SOLUTION: $OCC(Joe, Actor) \wedge \exists p\ (p \neq Actor) \wedge OCC(Joe, p)$

(c) All surgeons are doctors.
SOLUTION: $\forall p\ OCC(p, Surgeon) \Rightarrow OCC(p, Doctor)$

(d) Joe does not have a lawyer (i.e., he is not a customer of any lawyer).
SOLUTION: $\neg\exists p\ CUST(Joe, p) \wedge OCC(p, Lawyer)$.

(e) Emily has a boss who is a lawyer.
SOLUTION: $\exists p\ BOSS(p, Emily) \wedge OCC(p, Lawyer)$.

(f) There exists a lawyer all of whose customers are doctors.
SOLUTION: $\exists p\ OCC(p, Lawyer) \wedge [\forall q\ CUST(q, p) \Rightarrow OCC(q, Doctor)]$

(g) Every surgeon has a lawyer.
SOLUTION: $\forall p\ OCC(p, Surgeon) \Rightarrow [\exists q\ OCC(q, Lawyer) \wedge CUST(p, q)]$

7. [**First Order Logic**] Indicate which of the following Predicate Calculus sentences are correct representations of the corresponding English sentences. If not, explain why not.

		SOLUTION
1	$[\forall x$ PERSON$(x)] \Rightarrow [\exists y$ MOTHER$(y,x)]$ Everybody has a mother	No, the scope of the x quantifier is incorrect.
2	OLD(DOG(Fido)) Fido is an old dog	No, we can't have a predicate of a predicate.
3	DOG(Fido) $\wedge$ OLD(Fido) Fido is an old dog	YES
4	$\forall x$ [MATHEMATICAL-THEORY$(x) \Rightarrow x]$ All mathematical theories are true	No, you can't imply a variable.
5	$\exists s$ [ARISTOTLE-SAID(s) $\wedge \neg$TRUE$(s)]$ Aristotle told a lie	YES
6	$\neg\exists v\ \exists b$ [BUTCHER$(b) \wedge$ VEGETARIAN$(v)]$ There are no vegetarian butchers	No, the variables are different.
7	$\neg\exists b\ \exists d$ BUTCHER$(b) \wedge$ DOG$(d) \wedge$ OWNS(b,d) No butcher owns a dog	YES

8. [**Substitution**] Consider the following substitutions:
$s1 = \{y|x, z|A\}; s2 = \{x|C, z|y\}$
Perform the composition $s1s2$ and $s2s1$.

   SOLUTION:
   A combination of substitutions $s_i s_j$ is done as follows:

   (a) Apply s_j to the terms of s_i

   (b) Add any pairs of s_j having variables not in s_i

   - $s1s2$:
     Applying $s2$ to the terms of $s1$ (only applicable action is $x|C$): $\{y|C, z|A\}$
     Adding remaining variables of $s2$ (z is already in $s1$): $\{y|C, z|A, x|C\}$
   - $s2s1$:
     Applying $s1$ to the terms of $s2$ (only applicable action is $y|x$): $\{x|C, z|x\}$
     Adding remaining variables of $s1$ (z is already in $s2$): $\{x|C, z|x, y|x\}$

9. [**Unification**] For each of the following pairs of literals indicate whether or not they unify. If they do unify, show a most general unifier. If they do not unify, explain why not.

   (a) STRONGER(x, LexLuthor), STRONGER(LoisLane,y)
   SOLUTION: mgu = $\{x|$LoisLane,$y|$LexLuthor$\}$

   (b) HOLDING(x, Girlfriend(x)), HOLDING(Superman,y)
   SOLUTION: mgu = $\{x|$Superman,$y|$Girlfriend(Superman)$\}$

   (c) FLY(Girlfriend(x)), ¬FLY(LoisLane)
   SOLUTION: No, you can't unify a predicate with its negation, and you can't unify a constant with a function.

   (d) ¬LIKES(x,Enemy(x)), ¬LIKES(Enemy(y), y)
   SOLUTION: No, you can substitute $x|$Enemy(y), which yields ¬LIKES(Enemy(y),Enemy(Enemy(y))).
   However, you can't substitute a function that contains y for y itself.

10. [**Unification**] For each pair of atomic sentences, give the most general unifier if it exists. If they do not unify, explain why not.

    (a) $P(A, B, B), P(x, y, z)$.
    SOLUTION: $x|A, y|B, z|B$

    (b) $Q(y, G(A, B)), Q(G(x, x), y)$.
    SOLUTION: No unifier. These two sentences can firstly use $y|G(x, x)$, then they become $Q(G(x, x), G(A, B))$ and $Q(G(x, x), G(x, x))$. They cannot unify because x cannot bind to both A and B.

    (c) $OLDER(Father(y), y), OLDER(Father(x), John)$.
    SOLUTION: $y|John, x|John$.

    (d) $KNOWS(Father(y), y), KNOWS(x, x)$.
    SOLUTION: No unifier, because we first unify $x|Father(y)$, yielding $KNOWS(Father(y), y), KNOWS(Father(y), Father(y))$. Now, the second y cannot unify with $Father(y)$.

11. [**Resolution**] Resolve the following clauses, indicating clearly the literals you are resolving, their *mgu* and the resolvent:
$P(x, y) \vee \neg Q(A)$ and $W(x) \vee Q(x)$.

SOLUTION:
Literals: $\neg Q(A)$ and $Q(x)$
mgu: $\{x|A\}$
Resolvent: $P(A, y) \vee W(A)$

Note: if this question was part of an exercise where predicates must be converted to clauses, then the variables would have to be standardized apart, i.e.,
$P(x_1, y) \vee \neg Q(A)$ and $W(x_2) \vee Q(x_2)$.

12. [**Resolution**] Consider the following statements:

```
1. All hounds howl at night.
2. Anyone who has any cats will not have any mice.
3. Light sleepers do not have anything which howls at night.
4. John has a cat or a hound.
5. If John is a light sleeper, then John does not have any mice.
```

(a) Using only the predicates $HOUND$, $HOWL$, $HAVE$, CAT, $MOUSE$ and $LIGHTSLEEPER$, represent these statements as well formed formulas (wffs).

SOLUTION:
1. $\forall x\ (HOUND(x) \Rightarrow HOWL(x))$
2. $\forall p \forall y\ (HAVE(p, y) \wedge CAT(y) \Rightarrow \neg \exists z\ HAVE(p, z) \wedge MOUSE(z))$
3. $\forall p\ (LIGHTSLEEPER(p) \Rightarrow \neg \exists x\ (HAVE(p, x) \wedge HOWL(x)))$
4. $\exists w\ (HAVE(John, w) \wedge (CAT(w) \vee HOUND(w)))$
5. $LIGHTSLEEPER(John) \Rightarrow \neg \exists z\ (HAVE(John, z) \wedge MOUSE(z))$

(b) Convert Statements 1-4 to clauses.

SOLUTION:
1. $\neg HOUND(x1) \vee HOWL(x1)$
2. $\neg HAVE(p2, y2) \vee \neg CAT(y2) \vee \neg HAVE(p2, z2) \vee \neg MOUSE(z2)$
3. $\neg LIGHTSLEEPER(p3) \vee \neg HAVE(p3, x3) \vee \neg HOWL(x3)$
4.1 $HAVE(John, A)$
4.2 $CAT(A) \vee HOUND(A)$

(c) Use resolution to prove Statement 5.

SOLUTION:
Negate Statement 5 and convert to clause:
$\neg(LIGHTSLEEPER(John) \Rightarrow \neg \exists z\ (HAVE(John, z) \wedge MOUSE(z)))$
$\neg(\neg LIGHTSLEEPER(John) \vee \neg \exists z\ (HAVE(John, z) \wedge MOUSE(z)))$
$LIGHTSLEEPER(John) \wedge \exists z\ (HAVE(John, z) \wedge MOUSE(z))$
Skolemize: $LIGHTSLEEPER(John) \wedge HAVE(John, M) \wedge MOUSE(M)$
Clauses:
5.1 $LIGHTSLEEPER(John)$
5.2 $HAVE(John, M)$
5.3 $MOUSE(M)$

Applying resolution (please note that clauses can be re-used):
1. $\underline{\neg HOUND(x1)} \vee HOWL(x1)$ 4.2 $CAT(A) \vee \underline{HOUND(A)}$
substitution $\{x1|A\}$
6. $CAT(A) \vee HOWL(A)$
2. $\neg HAVE(p2, y2) \vee \neg CAT(y2) \vee \neg HAVE(p2, z2) \vee \underline{\neg MOUSE(z2)}$ 5.3 $\underline{MOUSE(M)}$
substitution $\{z2|M\}$
7. $\neg HAVE(p2, y2) \vee \neg CAT(y2) \vee \underline{\neg HAVE(p2, M)}$ 5.2 $\underline{HAVE(John, M)}$
substitution $\{p2|John\}$
8. $\neg HAVE(John, y2) \vee \underline{\neg CAT(y2)}$ 6. $\underline{CAT(A)} \vee HOWL(A)$
substitution $\{y2|A\}$
9. $\underline{\neg HAVE(John, A)} \vee HOWL(A)$ 4.1 $\underline{HAVE(John, A)}$
10. $\underline{HOWL(A)}$ 3. $\neg LIGHTSLEEPER(p3) \vee \neg HAVE(p3, x3) \vee \underline{\neg HOWL(x3)}$
substitution $\{x3|A\}$
11. $\neg LIGHTSLEEPER(p3) \vee \underline{\neg HAVE(p3, A)}$ 4.1 $\underline{HAVE(John, A)}$
substitution $\{p3|John\}$
12. $\underline{\neg LIGHTSLEEPER(John)}$ 5.1 $\underline{LIGHTSLEEPER(John)}$
NIL

13. [**Resolution**] From "Horses are animals", it follows that "The head of a horse is the head of an animal." Demonstrate that this inference is valid by carrying out the following steps:

(a) Translate the premise and the conclusion into a well formed formula (wff) in First Order Logic. Use three predicates: $HEADOF(h, x)$ (meaning "h is the head of x"), $HORSE(x)$, and $ANIMAL(x)$.
SOLUTION:
Premise: $\forall x\ HORSE(x) \Rightarrow ANIMAL(x)$
Conclusion: $\forall x \forall h\ HORSE(x) \wedge HEADOF(h, x) \Rightarrow \exists y\ ANIMAL(y) \wedge HEADOF(h, y)$

(b) Negate the conclusion, and convert the premise and the negated conclusion into CNF.
SOLUTION:
Premise: $\neg HORSE(x) \vee ANIMAL(x)$
Negated Conclusion:
$\neg(\forall x \forall h\ HORSE(x) \wedge HEADOF(h, x) \Rightarrow \exists y\ ANIMAL(y) \wedge HEADOF(h, y))$
$\neg(\forall x \forall h \neg(HORSE(x) \wedge HEADOF(h, x)) \vee \exists y\ ANIMAL(y) \wedge HEADOF(h, y))$
$(\exists x \exists h \neg\neg(HORSE(x) \wedge HEADOF(h, x)) \wedge \neg\exists y\ ANIMAL(y) \wedge HEADOF(h, y))$
$\exists x \exists h(HORSE(x) \wedge HEADOF(h, x)) \wedge \forall y \neg(ANIMAL(y) \wedge HEADOF(h, y))$
$\exists x \exists h(HORSE(x) \wedge HEADOF(h, x)) \wedge \forall y(\neg ANIMAL(y) \vee \neg HEADOF(h, y))$
Remove the universal quantifier and Skolemize:
$(HORSE(G) \wedge HEADOF(H, G)) \wedge (\neg ANIMAL(y) \vee \neg HEADOF(H, y))$
Clauses:
C1. $HORSE(G)$
C2. $HEADOF(H, G)$
C3. $\neg ANIMAL(y) \vee \neg HEADOF(H, y)$

(c) Use resolution to show that the conclusion follows from the premise.
SOLUTION:
Resolve C3 and C2 (substitution $y|G$): R1. $\neg ANIMAL(G)$
Resolve R1 with Premise (substitution $x|G$): R2. $\neg HORSE(G)$
Resolve R2 with C1: nil (contradiction)